A Logical Qubit with the Steane Code

Hackathon Project Report • Track: Quantum Error Correction

Project objective

Build and test a working Steane-code logical qubit in Qiskit, then demonstrate how error detection and correction can protect quantum information. The Steane [[7,1,3]] code encodes one logical qubit into seven physical qubits and can correct any single-qubit Pauli error: X (bit flip), Z (phase flip), or Y (both).

Why this matters

Physical qubits are susceptible to noise. Quantum error correction stores information across an entangled multi-qubit code space and uses stabilizer measurements to learn about errors without directly measuring the logical state. The Steane code is a useful demonstration because it is based on the classical [7,4,3] Hamming code and has six stabilizer generators.

What was implemented

Stage	Implementation / evidence
1. Encode	Prepared the seven-qubit logical-zero state as an equal superposition of the eight Hamming-code basis words.
2. Validate stabilizers	Checked all six X- and Z-type stabilizer expectations; the notebook output reports +1 for all six.
3. Inject an error	Introduced a known single-qubit Pauli error and measured stabilizer syndromes using six ancilla qubits.
4. Correct conditionally	Used Qiskit classical registers and if_test-controlled X/Z corrections, then measured the data qubits.
5. Test coverage	Tested X, Y and Z on each of seven data qubits: 21/21 ideal single-error tests passed (100 shots per test).

Main result: the ideal Qiskit correction tests passed for all 21 single-qubit X/Y/Z errors. A separate Monte Carlo benchmark studied logical failure under independent depolarizing Pauli noise with perfect recovery.

How the Steane Code Works

The code uses three parity-check supports from the Hamming code. Each support defines one X-type stabilizer and one Z-type stabilizer, giving six generators in total. The supports below are written in the notebook's displayed order; the code reverses the string index when mapping bits to physical qubits.

Generator	Pauli type	Support string
S1	X	0011110
S2	Z	0011110
S3	X	0101101
S4	Z	0101101
S5	X	1001011
S6	Z	1001011

Syndrome extraction and correction

Three Z-type checks reveal the X component of an error; three X-type checks reveal the Z component. A non-zero syndrome identifies the affected qubit under the single-error assumption. The circuit uses six ancillas to measure the checks, classical registers to store the syndromes, and Qiskit's if_test to conditionally apply corrections.

Ideal validation results

Test	Notebook result	Interpretation
Stabilizer expectations	All six = +1	Prepared state is in the expected stabilizer space.
Single-qubit Pauli tests	21 / 21 PASS	Each X, Y or Z error on q0–q6 was detected/corrected in the ideal circuit test.
Shots per test	100	The ideal test was repeated 100 times per injected error.

What the tests do—and do not—prove

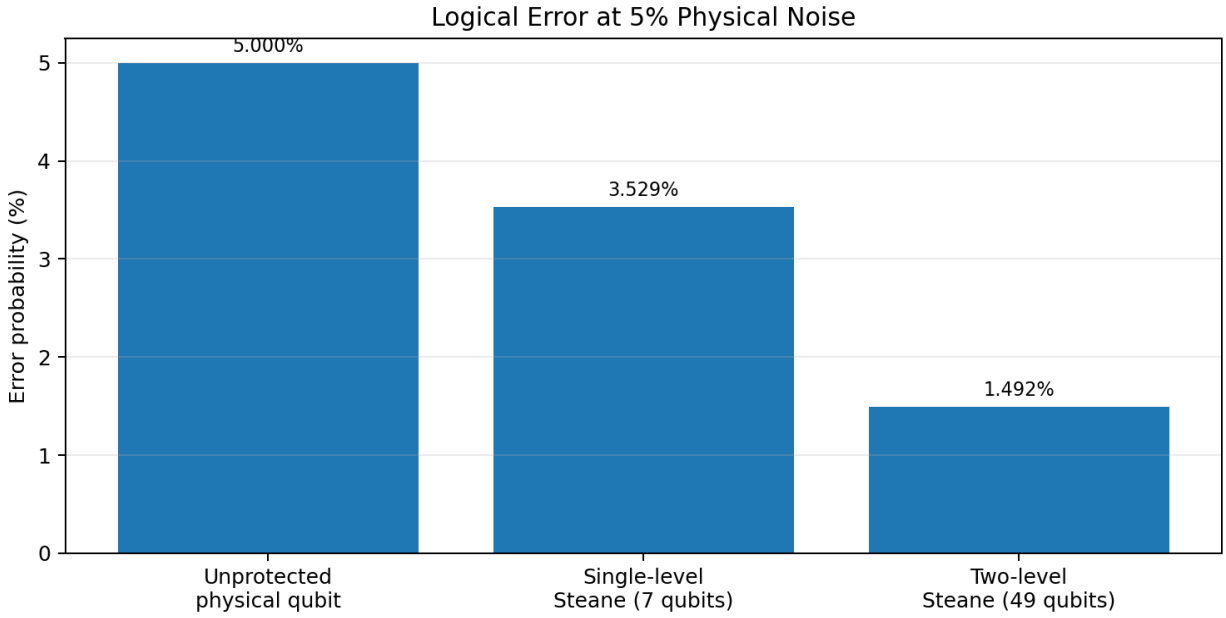
The 21/21 result is evidence that the ideal syndrome-and-correction circuit handles all single-qubit Pauli error types for the tested encoded logical-zero state. Computational-basis codeword membership alone is not a full logical-fidelity measurement for arbitrary superpositions. The notebook also explores gate-noise diagnostics; code-space success percentages should not be described as logical fidelity.

Performance Benchmark

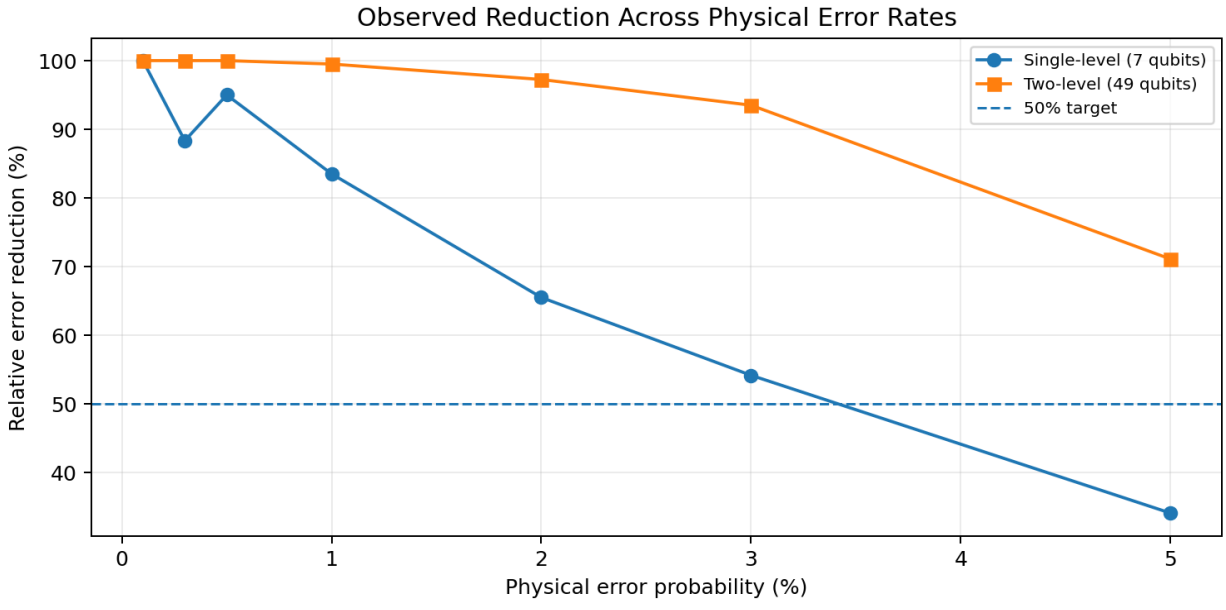
A separate phenomenological benchmark applies independent depolarizing Pauli noise to physical qubits. Each qubit has total error probability p, with X, Y and Z each occurring with probability p/3. Syndrome extraction and recovery are assumed perfect. At p = 5%, the notebook's 100,000-trial comparison reported the following results:

Encoding	Physical qubits	Failures / trials	Logical error	Relative reduction
Unprotected	1	—	5.000%	0.00%
Single-level Steane	7	3,529 / 100,000	3.529%	29.42%
Two-level Steane	49	1,492 / 100,000	1.492%	70.16%

At 5% physical noise, the two-level result gives a relative reduction of 70.16% (approximate 95% interval: 68.66%–71.66%). The single-level reduction is 29.42% (approximate 95% interval: 27.13%–31.71%). These intervals quantify Monte Carlo sampling uncertainty only.



Observed trend across noise rates



The sweep graph uses the notebook's recorded 20,000-trial estimates from 0.1% to 5% physical noise. At very low error rates, zero observed failures in a finite sample does not prove the true logical error is exactly zero. Monte Carlo values can fluctuate, especially when few failures occur.

Conclusion and limitations

The project demonstrates the main workflow of a Steane logical qubit: encoding, stabilizer checks, syndrome extraction, conditional recovery, and ideal single-error testing. The recorded ideal tests passed 21/21 cases. The separate 49-qubit benchmark reported 1.492% logical failure at 5% physical noise, corresponding to a 70.16% reduction compared with a 5% unprotected error rate.

Scope: the 70.16% benchmark is an idealized simulation, not a hardware result. It assumes perfect recovery and excludes faulty syndrome measurements, ancilla preparation, and gate errors from that benchmark. A full fault-tolerant circuit study would need to model these sources and assess logical-state fidelity.

Software used in the notebook: Qiskit, Qiskit Aer, NumPy, and Matplotlib. The benchmark results are Monte Carlo estimates; rerunning with a different seed or shot count may change the estimates slightly.